

```
In [0]: df = spark.table("test.default.sample_superstore")

display(df)
```

Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country	City	State	Postal Code
1	CA-2016-152156	2016-11-08	2016-11-11	Second Class	CG-12520	Claire Gute	Consumer	United States	Henderson	Kentucky	42
2	CA-2016-152156	2016-11-08	2016-11-11	Second Class	CG-12520	Claire Gute	Consumer	United States	Henderson	Kentucky	42
3	CA-2016-138688	2016-06-12	2016-06-16	Second Class	DV-13045	Darrin Van Huff	Corporate	United States	Los Angeles	California	90
4	US-2015-108966	2015-10-11	2015-10-18	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States	Fort Lauderdale	Florida	33
5	US-2015-108966	2015-10-11	2015-10-18	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States	Fort Lauderdale	Florida	33
6	CA-2014-115812	2014-06-09	2014-06-14	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90
7	CA-2014-115812	2014-06-09	2014-06-14	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90
8	CA-2014-115812	2014-06-09	2014-06-14	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90
9	CA-2014-115812	2014-06-09	2014-06-14	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90
10	CA-2014-115812	2014-06-09	2014-06-14	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90
11	CA-2014-115812	2014-06-09	2014-06-14	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90
12	CA-2014-115812	2014-06-09	2014-06-14	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90
13	CA-2017-114412	2017-04-15	2017-04-20	Standard Class	AA-10480	Andrew Allen	Consumer	United States	Concord	North Carolina	28
14	CA-2016-161389	2016-12-05	2016-12-10	Standard Class	IM-15070	Irene Maddox	Consumer	United States	Seattle	Washington	98
15	US-2015-118983	2015-11-22	2015-11-26	Standard Class	HP-14815	Harold Pawlan	Home Office	United States	Fort Worth	Texas	76
16	US-2015-118983	2015-11-22	2015-11-26	Standard Class	HP-14815	Harold Pawlan	Home Office	United States	Fort Worth	Texas	76
17	CA-2014-105893	2014-11-11	2014-11-18	Standard Class	PK-19075	Pete Kriz	Consumer	United States	Madison	Wisconsin	53
18	CA-2014-167164	2014-05-13	2014-05-15	Second Class	AG-10270	Alejandro Grove	Consumer	United States	West Jordan	Utah	84

19	CA-2014-143336	2014-08-27	2014-09-01	Second Class	ZD-21925	Zuschuss Donatelli	Consumer	United States	San Francisco	California	94
20	CA-2014-143336	2014-08-27	2014-09-01	Second Class	ZD-21925	Zuschuss Donatelli	Consumer	United States	San Francisco	California	94

[Showing first 20 of 7464 rows — full output available in the notebook/GitHub repo]

Showing the first 7464 rows.

```
In [0]: print(df.count())

9994
```

```
In [0]: df.printSchema()

root
|-- Row ID: long (nullable = true)
|-- Order ID: string (nullable = true)
|-- Order Date: date (nullable = true)
|-- Ship Date: date (nullable = true)
|-- Ship Mode: string (nullable = true)
|-- Customer ID: string (nullable = true)
|-- Customer Name: string (nullable = true)
|-- Segment: string (nullable = true)
|-- Country: string (nullable = true)
|-- City: string (nullable = true)
|-- State: string (nullable = true)
|-- Postal Code: long (nullable = true)
|-- Region: string (nullable = true)
|-- Product ID: string (nullable = true)
|-- Category: string (nullable = true)
|-- Sub-Category: string (nullable = true)
|-- Product Name: string (nullable = true)
|-- Sales: double (nullable = true)
|-- Quantity: long (nullable = true)
|-- Discount: double (nullable = true)
|-- Profit: double (nullable = true)
```

```
In [0]: %sql
DESCRIBE DETAIL test.default.sample_superstore;
```

format	id	name	description	location	createdAt	lastModifie
delta	e9f8be9d-feff-407d-8507-9e84d75b83b2	test.default.sample_superstore	Created by the file upload UI		2026-07-12T18:58:55.179Z	2026-07-12T18:59:02.000Z

```
In [0]: df.columns
```

```
Out[0]: ['Row ID',
        'Order ID',
        'Order Date',
        'Ship Date',
        'Ship Mode',
        'Customer ID',
        'Customer Name',
        'Segment',
        'Country',
        'City',
        'State',
        'Postal Code',
        'Region',
        'Product ID',
        'Category',
        'Sub-Category',
        'Product Name',
        'Sales',
        'Quantity',
        'Discount',
        'Profit']
```

```
In [0]: from pyspark.sql import Row

updates = [
    Row(
        **{
            "Row ID": 1,
            "Order ID": "CA-2016-152156",
            "Sales": 999.99,
            "Quantity": 5,
            "Profit": 250.50
        }
    ),
    Row(
        **{
            "Row ID": 99999,
            "Order ID": "CA-2026-999999",
            "Sales": 500.00,
            "Quantity": 2,
            "Profit": 120.00
        }
    )
]

updates_df = spark.createDataFrame(updates)

display(updates_df)
```

```
In [0]: from pyspark.sql.types import StructType, StructField, IntegerType, StringType, DateType, DoubleType
        from datetime import date

schema = StructType([
    StructField("Row ID", IntegerType(), True),
    StructField("Order ID", StringType(), True),
    StructField("Order Date", DateType(), True),
    StructField("Ship Date", DateType(), True),
    StructField("Ship Mode", StringType(), True),
    StructField("Customer ID", StringType(), True),
    StructField("Customer Name", StringType(), True),
    StructField("Segment", StringType(), True),
    StructField("Country", StringType(), True),
    StructField("City", StringType(), True),
    StructField("State", StringType(), True),
    StructField("Postal Code", IntegerType(), True),
    StructField("Region", StringType(), True),
    StructField("Product ID", StringType(), True),
    StructField("Category", StringType(), True),
    StructField("Sub-Category", StringType(), True),
    StructField("Product Name", StringType(), True),
    StructField("Sales", DoubleType(), True),
    StructField("Quantity", IntegerType(), True),
    StructField("Discount", DoubleType(), True),
    StructField("Profit", DoubleType(), True)
])
```

```

data = [
(
1,
"CA-2016-152156",
date(2016,11,8),
date(2016,11,11),
"Second Class",
"CG-12520",
"Claire Gute",
"Consumer",
"United States",
"Henderson",
"Kentucky",
42420,
"South",
"FUR-B0-10001798",
"Furniture",
"Bookcases",
"Bush Somerset Collection Bookcase",
999.99,
5,
0.0,
250.50
),
(
99999,
"CA-2026-999999",
date(2026,7,13),
date(2026,7,15),
"Standard Class",
"CUST-999",
"Rishi Jain",
"Consumer",
"United States",
"Jaipur",
"Rajasthan",
302017,
"West",
"OFF-NEW-001",
"Office Supplies",
"Binders",
"Assignment Product",
500.00,
2,
0.0,
120.00
)
]

updates_df = spark.createDataFrame(data, schema)

display(updates_df)

```

Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country	City	State	Postal Code
1	CA-2016-152156	2016-11-08	2016-11-11	Second Class	CG-12520	Claire Gute	Consumer	United States	Henderson	Kentucky	42420
99999	CA-2026-999999	2026-07-13	2026-07-15	Standard Class	CUST-999	Rishi Jain	Consumer	United States	Jaipur	Rajasthan	302017

```
In [0]: from delta.tables import DeltaTable
```

```
In [0]: deltaTable = DeltaTable.forName(spark, "test.default.sample_superstore")
```

```
In [0]: (deltaTable.alias("target")
.merge(
updates_df.alias("source"),
"target.`Order ID` = source.`Order ID`"
)
.whenMatchedUpdate(set={

```

```

    "Sales": "source.Sales",
    "Quantity": "source.Quantity",
    "Profit": "source.Profit"
  })
  .whenNotMatchedInsertAll()
  .execute()
)

```

```
Out[0]: DataFrame[num_affected_rows: bigint, num_updated_rows: bigint, num_deleted_rows: bigint, num_inserted_rows: bigint]
```

```
In [0]: display(spark.table("test.default.sample_superstore"))
```

Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country	City	State	Postal Code
3	CA-2016-138688	2016-06-12	2016-06-16	Second Class	DV-13045	Darrin Van Huff	Corporate	United States	Los Angeles	California	90
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20	CA-2014-143336	2014-08-27	2014-09-01	Second Class	ZD-21925	Zuschuss Donatelli	Consumer	United States	San Francisco	California	94
21	CA-2014-143336	2014-08-27	2014-09-01	Second Class	ZD-21925	Zuschuss Donatelli	Consumer	United States	San Francisco	California	94
22	CA-2016-137330	2016-12-09	2016-12-13	Standard Class	KB-16585	Ken Black	Corporate	United States	Fremont	Nebraska	68

[Showing first 20 of 7464 rows — full output available in the notebook/GitHub repo]

Showing the first 7464 rows.

In [0]: %sql
SELECT *
FROM test.default.sample_superstore
WHERE `Order ID` IN ('CA-2016-152156', 'CA-2026-999999');

Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country	City	State	Pos C
2	CA-2016-152156	2016-11-08	2016-11-11	Second Class	CG-12520	Claire Gute	Consumer	United States	Henderson	Kentucky	42
1	CA-2016-152156	2016-11-08	2016-11-11	Second Class	CG-12520	Claire Gute	Consumer	United States	Henderson	Kentucky	42
99999	CA-2026-999999	2026-07-13	2026-07-15	Standard Class	CUST-999	Rishi Jain	Consumer	United States	Jaipur	Rajasthan	302